

TAMIL NADU STATE BOARD - STANDARD 9 MATHEMATICS

CHAPTER 5: COORDINATE GEOMETRY

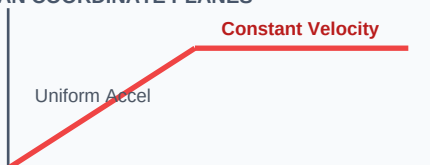
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Mathematics
Chapter Name:	Coordinate Geometry

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

UNIT ROADMAP: CARTESIAN COORDINATE PLANES



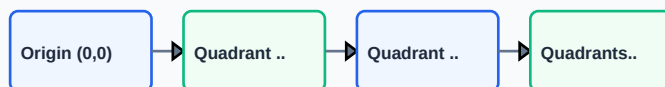
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Cartesian Plane	A coordinate system defined by a horizontal X-axis and vertical Y-axis intersecting at the origin (0, 0).
Distance Formula	Analytical formula used to calculate the length of the straight line segment between two coordinate points.
Section Formula	Finds the coordinates of a point that divides a line segment joining two points in a given ratio $m:n$.
Centroid	The point where the three medians of a triangle intersect. It divides each median in the ratio 2:1.

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

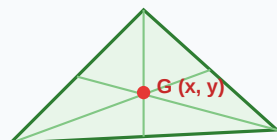
GLOSSARY: CARTESIAN PLANE QUADRANTS



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

FORMULA: TRIANGLE CENTROID G COORDINATES



Essential Formulas, Laws & Identities:

- Distance: $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$
- Midpoint: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
- Section Formula (Internal): $P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$
- Centroid coordinates: $G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find the distance between the points (3, -2) and (-1, 1).

Step-by-step Solution:

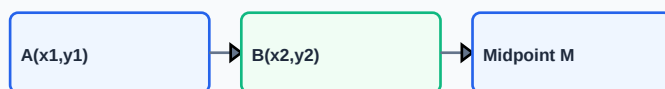
Apply distance formula: $d = \sqrt{(-1 - 3)^2 + (1 - (-2))^2}$ $d = \sqrt{(-4)^2 + (3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$ units.

Example 2: Find the centroid of the triangle with vertices (1, 3), (2, 7), and (12, -1).

Step-by-step Solution:

$G = ((1+2+12)/3, (3+7-1)/3) = (15/3, 9/3) = (5, 3)$. So centroid is (5, 3).

WORKED: MIDPOINT FORMULA COORDINATE STEP



HOTS (Higher Order Thinking Skills) Challenge

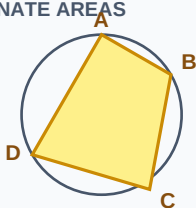
Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Find the coordinates of the point which divides the line segment joining (3, 5) and (8, 10) internally in the ratio 2:3.

Analytical Solution Strategy:

Given: $x_1=3$, $y_1=5$, $x_2=8$, $y_2=10$, $m=2$, $n=3$. $x = \frac{2 \cdot 8 + 3 \cdot 3}{2+3} = \frac{16+9}{5} = \frac{25}{5} = 5$. $y = \frac{2 \cdot 10 + 3 \cdot 5}{2+3} = \frac{20+15}{5} = \frac{35}{5} = 7$. Point coordinates are (5, 7).

HOTS: POLYGON BOUNDARY COORDINATE AREAS



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Calculate the midpoint of the segment joining $(-4, 2)$ and $(6, -8)$.

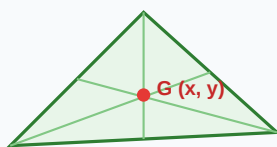
Q2 (2-Mark): Show that the points $(1, 1)$, $(3, 4)$, and $(5, 7)$ are collinear using distance formula.

Q3 (2-Mark): Find the distance of the point $(6, 8)$ from the origin.

Q4 (5-Mark): If the centroid of a triangle is $(2, 3)$ and two vertices are $(1, 2)$ and $(3, 5)$, find the third vertex.

Q5 (5-Mark): Calculate the area of the triangle with vertices $(0, 0)$, $(3, 0)$, and $(0, 4)$.

PRACTICE: DIVISION OF LINE SEGMENTS INTERNAL



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Mapping & Navigation

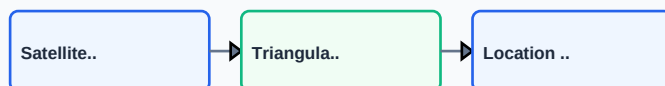
GPS coordinates represent latitude and longitude on a spherical model mapped onto a coordinate plane for path-finding.

Application Case: Computer Games

Game engines represent 2D/3D sprites as coordinate positions and translate/rotate them using matrix geometry arithmetic.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: COORDINATE SYSTEMS IN GPS NAVIGATION



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

REVISION: COORDINATE GEOMETRY FLOWCHART

